

2017 Ph H2 Q6

Section: Our Dynamic Universe

Topic: The Expanding Universe

The spectrum of light from a distant galaxy is observed on Earth.
A particular spectral line has:

- **Wavelength in lab:** $\lambda_0 = 486 \text{ nm}$
- **Observed wavelength:** $\lambda = 510 \text{ nm}$

(a)

State the name of the effect that causes this change in wavelength.

✔ This is called **redshift**.

| Light from the galaxy is **stretched** to longer wavelengths as the galaxy moves **away**.

(b)

Calculate the redshift, z , for this spectral line.

Use the redshift formula:

$$z = \frac{\lambda - \lambda_0}{\lambda_0} = \frac{510 - 486}{486} = \frac{24}{486} = 0.0494$$

✔ *Revision tip:*

Redshift is a **ratio**, not a unit.

If the wavelength increases, $z > 0 \rightarrow$ galaxy is **receding**.

(c)

Explain how the observation of redshift in the light from galaxies supports the theory that the Universe is expanding.

✔ Redshift shows that **most galaxies are moving away from us**.

The greater the redshift, the faster a galaxy is receding.

This supports the idea that:

| All galaxies are moving away from each other, which implies the Universe is **expanding** — consistent with the **Big Bang theory**.

✔ *Revision tip:*

The farther the galaxy, the **greater** the redshift — this is **Hubble’s Law** in action!